

We want to show that if we have an equation of the form $f(x, y) = c$ (where f is differentiable with a continuous derivative as a function of x and y) then in an interval around any non-degenerate (we will see what this means) point, the curve set can locally be written as $y = f(x)$ with f a continuously differentiable function. Note that there may be easier ways to prove this, but I decided to stick with this inefficient proof path from wikipedia because it turns out a lot of the preliminary material will be incredibly useful in upcoming levels.

Definition 1. A function is Lipschitz continuous if we always satisfy that $|f(x) - f(y)| \leq k|x - y|$ for some k . This intuitively means its slope is never greater than k , however it need not be differentiable, it just can't ever be "vertical" or approach being vertical.

Definition 2. A family of functions F is equicontinuous at a point x_0 if for every $\varepsilon > 0$ there exists a $\delta > 0$ such that for all functions f in F ,

$$|x_0 - x| < \delta \Rightarrow |f(x_0) - f(x)| < \varepsilon$$

A family of functions is globally equicontinuous if for every ε there is a δ such that for any function in the family and any 2 points x_0, x less than δ apart we have the implication

$$|x_0 - x| < \delta \Rightarrow |f(x_0) - f(x)| < \varepsilon$$

Definition 3. Partial derivatives

We can have functions of multiple variables, like $f : \mathbb{R}^m \rightarrow \mathbb{R}^n$. As an example, suppose $z = x^2 + y^3$, then we can sketch this using a contour plot, kind of like elevation maps, where we show lines on the x, y -plane corresponding to when $x^2 + y^3$ is constant. Here is that example (Figure 1):

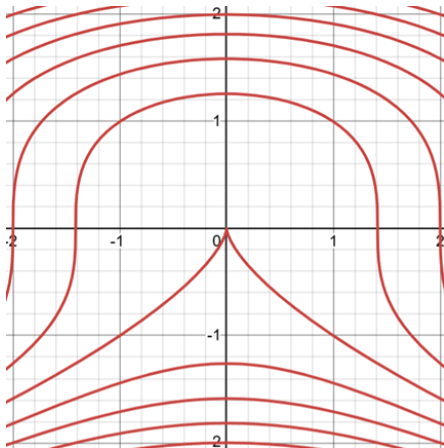


Figure 1

However, if I imagine this as a 3d graph of a surface with height equal to $x^2 + y^3$, then if I pick a point on this graph and try to find the slope, I have a problem that the slope depends on the direction. Therefore I write $\frac{\partial z}{\partial x}$ for the slope as I move in just the x direction and hold y constant. This is called a partial derivative. In this example, that is $2x$, because we differentiate $x^2 + y^3$ and the y^3 differentiates to 0 since it is held constant. We put a little thingy in the corner like this to show what's being held constant, as shown below.

$$\frac{\partial z}{\partial x} \Big|_y$$

This is the simple case of a partial derivative, where we hold all but 1 variable constant. If f is a function of x, y and z , then we could hold one variable constant and add another constraint. For example, suppose we are on the surface $x^2 + y^3 + z^4 = 1$ and y is constrained to be constant. Then z becomes a function of x along the constrained surface, and on this curve $\frac{d}{dx}(x^2 + y^3 + z^4) = 2x + 4z^3 \frac{dz}{dx}$ since y is constant, however $\frac{\partial(x^2 + y^3 + z^4)}{\partial x} \Big|_z = 2x + 3y^2 \frac{dy}{dx}$.

Formally, for example, if z is a function of x and y , then $\frac{\partial z}{\partial x} \Big|_y$ is defined by $\lim_{h \rightarrow 0} \frac{z(x+h, y) - z(x, y)}{h}$.

Example. $f(x, y) = x^2 + y^3 + e^{xy^2}$. As a shorthand for the partial derivative with respect to x we often instead write f_x . Since y is treated as constant here, we get that $f_x = 2x + y^2 e^{xy^2}$, and $f_y = 3y^2 + 2xy e^{xy^2}$.

Lemma 1. (Piecewise linear approximation) For any continuous real function defined on a closed rectangle $[a, b] \times [c, d]$ we can define a sequence of Lipschitz continuous functions that converges uniformly to our function.

Proof. Fix $\varepsilon > 0$. We will use the known fact (Level 6.2) that our function is uniformly continuous since it is continuous on a closed interval. So let δ be such that $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \frac{\varepsilon}{2}$. Now what we will do is split our rectangle into rectangles with long side shorter than $\frac{\delta}{\sqrt{2}}$ (actually we just need that the longest diagonal is shorter than δ). Now in this rectangle we will define our function g to coincide with f at the corners and then split each rectangle into 2 triangles along the diagonal and have our function change linearly within each triangle. Now for any t in this rectangle with x_0 one of the rectangle's corners and y_0 its opposite corner, $|x_0 - t| < \delta \Rightarrow |f(x_0) - f(t)| < \varepsilon$, and also $|y_0 - t| < \delta \Rightarrow |f(y_0) - f(t)| < \varepsilon$. Therefore what we have is that $f(x_0)$ and $f(y_0)$ are within a band of width 2ε around $f(t)$, so thus $g(t)$ which is between $g(x_0)$ and $g(y_0)$ is in that band, so we must have $|g(t) - f(t)| < \varepsilon$. Therefore if we let $\varepsilon_n \rightarrow 0$, we will get uniform convergence, and all of these individual functions are Lipschitz continuous because of how we defined them: Their slopes are bounded by $\sqrt{g_x^2 + g_y^2}$, where these slopes are finite because they are the finite change in the function across the rectangle divided by the non-zero width of the rectangle.

□

Lemma 2. A closed and bounded set in $\mathbb{R}^n$ (n-dimensional space) has the property that if you cover the set by a bunch of open sets, then you must be able to find a finite subset of those open sets that still covers the original set.

Proof. Ok so suppose we have a closed bounded subset of $\mathbb{R}^n$ and suppose it is contained in a cube of side length M (possible by boundedness) and suppose we cover this closed bounded subset by open sets, then we want to show that we can get a finite subcover. So here is what we do: We take this cube and take our set S and split it into 2^n subcubes each of side length $\frac{M}{2}$, then one of these cubes intersected with S is a closed set where the part of our open cover that covers that set has no finite subcover – if all 2^n such intersections had a finite subcover so would the whole thing. Now take this subcube which we know exists and split it into subcubes again. When we iterate this, it is easy to see that we will converge to a single point in S (S is closed so it contains its limit points). Now this point contains an open set as part of the cover, but recall that open means every point has some small open ball centered at that point that is in the set. So if we take a subcube in the sequence that is much smaller than this open ball, it will be contained entirely within this open ball. So clearly, S intersected with this subcube has a finite subcover from the original open cover – in fact it is not only finite but covered by a single open set. This is a contradiction, so we have now proven the theorem.

□

Lemma 3. Suppose we have a sequence of continuous functions converging uniformly, then the limit is continuous.

Proof. Fix $\varepsilon > 0$. Pick N such that for all $n > N$, and all t $|f_n(t) - f(t)| < \frac{\varepsilon}{3}$. By continuity, for each x we have that

$$|x - y| < \delta \Rightarrow |f_N(x) - f_N(y)| < \frac{\varepsilon}{3}$$

for some δ . Then, by the triangle inequality,

$$|x - y| < \delta \Rightarrow |f(x) - f(y)| \leq |f(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f(y) - f_N(y)| < \varepsilon.$$

□

Lemma 4. (Arzela-Ascoli theorem specialized to real functions on 2D closed sets) Let a sequence of functions f_n be uniformly bounded on a closed bounded set (For our purposes, we will prove this for 1D or 2D sets) and equicontinuous, then there exists a subsequence f_{n_k} that converges uniformly to a continuous function f .

Proof. Enumerate the rational points in $\mathbb{R}^2$. To do this, first enumerate the rationals, i.e. write them out in a list so that we have a one-to-one mapping between the positive integers and the rational numbers. There are many ways to do this, but one way is to go around like this image below where the x is the numerator and the y is the denominator, but exclude duplicates or 0 denominators.

Figure 2 shows a visual idea of how we will do this

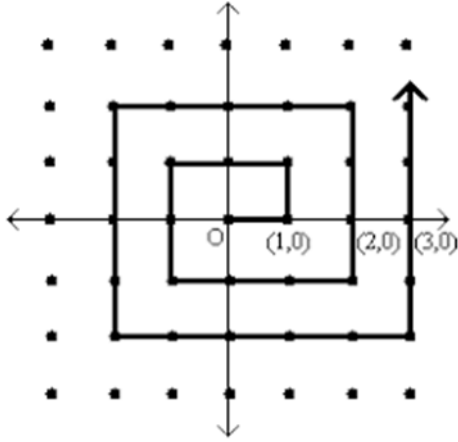


Figure 2

Then we know from level 6 technical results that the Cartesian product of two listable sets is listable, in fact by a similar diagram to the image above.

Now we filter this enumeration so we only have the rational points in our set. So we have a list of all the rational points in our interval. Call this enumeration $x_1, x_2, \dots$

Since f_n has a uniform bound M , there is a sequence $f_{n_{1,k}}$ such that $f_{n_{1,k}}(x_1)$ converges pointwise by Bolzano-Weierstrass (Level 4). We can find a further subsequence $f_{n_{2,k}}$ of this such that $f_{n_{2,k}}(x_2)$ also converges. We can get an infinite chain of subsequences this way. Now we want to form a sequence of functions g_k defined by $g_k = f_{n_{k,k}}$. By construction, this converges at every rational point. Therefore, given any $\varepsilon > 0$ and any rational point x_k , we can find an integer N such that for all $n, m > N$, we have that $|g_n(x_k) - g_m(x_k)| < \frac{\varepsilon}{3}$. We're making progress.

Since the family F is equicontinuous, there must be an open interval around x_k such that for any s and t in that open interval, $|f(s) - f(t)| < \frac{\varepsilon}{3}$ for all f in our family of functions. The open intervals each cover a point of length δ around the rational points for some universal δ depending only on our family of functions.

Subtlety: It is possible to cover all rational points by open intervals but not cover the whole real line if we cover x_k by an interval of length $\frac{1}{2^k}$, as then the whole cover will have length at most 1. Although this fact is counterintuitive when you first see it, it does mean the universal δ is crucial.

Doing this for all x_k gives a covering of our set using open sets. Then extract a finite subcover. Then there exists an integer K such that each of these open sets in our subcover contains one of the first K rationals in our list, otherwise our list would be missing every rational in that interval. Finally, for any t in our interval, there are j and $k < K$ such that t and x_k belong to the same interval U_j . For this choice of k , we have, by the triangle inequality:

$$|g_n(t) - g_m(t)| \leq |g_n(t) - g_n(x_k)| + |g_m(x_k) - g_n(x_k)| + |g_m(t) - g_m(x_k)|$$

Where we pick n and m to be at least as large as N which is at least large enough such that for all k from 1 to K we have the above inequality $|g_n(x_k) - g_m(x_k)| < \frac{\varepsilon}{3}$, and also from how we defined the U 's, it is always the case that

$$|g_n(t) - g_n(x_k)|, |g_m(t) - g_m(x_k)| < \frac{\varepsilon}{3}$$

Therefore what we have is that for any t and fixed ε , $|g_n(t) - g_m(t)| \leq \varepsilon$ for m and n large enough. Therefore at each t , we see that the functions value is forced into an arbitrarily small band and thus we have pointwise convergence. We define f to be this pointwise limit, and then $|g_n(t) - g_m(t)| \leq \varepsilon$. Now letting m go to infinity and taking a pointwise limit, $|g_n(t) - f(t)| \leq \varepsilon$, so our sequence of functions converges uniformly. Since ε was arbitrary, we can say that, for example, $|g_n(t) - f(t)| \leq \frac{\varepsilon}{2} < \varepsilon$ so the inequality is strict. Then conclude since uniform limits of continuous functions are continuous.

□

Lemma 5. (Peano existence theorem) Let $f(x, y)$ be continuous on an open set D around (x_0, y_0) , then the differential equation $y'(x) = f(x, y)$ with initial condition (x_0, y_0) has a solution in a neighbourhood about that point that is not

necessarily unique. (It is unique under another mild condition, when f is Lipschitz continuous turns out to be sufficient, but we do not need that so we will just do the existence theorem for now).

Proof. By replacing y with $y - y_0$ and similarly for x , we can assume that the initial condition is that we must pass through the origin. Since D is open, define a closed rectangle $R := [-x_1, x_1] \times [-y_1, y_1]$ contained in D . On R , the extreme value theorem implies that there exists $C > 0$ such that $\sup_R |f| \leq C < \infty$. Now by piecewise linear approximation, pick a sequence of Lipschitz continuous functions f_n converging uniformly to f with $\sup_R |f_k| \leq 2C < \infty$. We define the Picard iterations $y_{k,n} : I = [-t_2, t_2] \rightarrow \mathbb{R}$ where we set that $t_2 = \min(x_1, \frac{y_1}{2C})$, as follows:

In the work below if $x < 0$ then you should consider $t \in [0, x]$ to instead mean $t \in [x, 0]$.

$y_{k,0}(t) = 0$ and $y_{k,n+1}(x) = \int_0^x f_k(t, y_{k,n}(t)) dt$. They are well defined by induction as we have that

$$|y_{k,n+1}(x)| \leq \left| \int_0^x |f_k(t, y_{k,n}(t))| dt \right| \leq |x| \sup_R |f_k| \leq t_2 2C \leq y_1$$

We need the absolute value outside the integral in case $x < 0$.

and thus $(t, y_{k,n}(t))$ is within the domain of f_k . Also, by the triangle inequality for integrals,

$$|y_{k,n+1}(x) - y_{k,n}(x)| \leq \left| \int_0^x |f_k(t, y_{k,n}(t)) - f_k(t, y_{k,n-1}(t))| dt \right| \leq L_k \left| \int_0^x |(t, y_{k,n}(t)) - (t, y_{k,n-1}(t))| dt \right|$$

Where for each k , an L_k exists by the Lipschitz condition.

Now define

$$M_{k,n}(x) = \sup_{t \in [0, x]} |y_{k,n+1}(t) - y_{k,n}(t)| \leq L_k \left| \int_0^x M_{k,n-1}(t) dt \right|$$

We also have that

$$M_{k,0}(x) = \sup_{t \in [0, x]} |y_{k,1}(t) - y_{k,0}(t)| = \sup_{t \in [0, x]} |y_{k,1}(t)| \leq \left| \int_0^x |f_k(t, 0)| dt \right| \leq 2C|x|$$

Now we will prove by induction what we have the following bound for x in I for which we just proved the base case:

$$M_{k,n}(x) \leq \frac{(2CL_k^n |x|^{n+1})}{(n+1)!}$$

Let's do the induction step. Suppose this is true, then

$$M_{k,n+1}(t) \leq L_k \left| \int_0^x M_{k,n}(t) dt \right| \leq L_k \left| \int_0^x \frac{(2CL_k^n |x|^{n+1})}{(n+1)!} dt \right| \leq 2CL_k^{n+1} \left| \int_0^x \frac{(|x|^{n+1})}{(n+1)!} dt \right| = \frac{(2CL_k^{n+1} |x|^{n+2})}{(n+2)!}$$

As required. Crucially, this tends to 0 as n goes to infinity for all fixed x .

Now for x and x' in I , $|y_{k,n+1}(x') - y_{k,n+1}(x)| \leq \left| \int_x^{x'} |f_k(t, y_{k,n}(t))| dt \right| \leq 2C|x' - x|$, and thus since this always holds, the family of functions $y_{k,n}$ is equicontinuous: For ε given pick $\delta = \frac{\varepsilon}{2C}$. Therefore by Arzela Ascoli, for each k , there is a subsequence y_{k,a_n} converging uniformly to a continuous function y_k .

$$\left| y_{k,a_n}(x) - \int_0^x f_k(t, y_{k,a_n}(t)) dt \right| = |y_{k,a_n}(x) - y_{k,a_n+1}(x)| \leq M_{k,a_n}(x) \rightarrow 0$$

Thus, for each fixed x , we conclude $y_k(x) = \int_0^x f_k(t, y_k(t)) dt$ since the limit of $y_{k,a_n}(x)$ must coincide according to the inequality above, and the integral approaches $\int_0^x f_k(t, y_k(t)) dt$ since f is continuous so we can pass the y limit through it and then we are bounded by $2C$ so we can use dominated convergence (level 6 technical results) to pass the limit through

the integral. Since each $y_{k,n}$ has a uniform bound y_1 , so does y_k (the limit of those). Now, by the triangle inequality for integrals,

$$|y_k(x) - y_k(x')| \leq \left| \int_{x'}^x |f_k(t, y_k(t))| dt \right| \leq 2C|x - x'|$$

So y_k is equicontinuous so it has a subsequence k_n that converges uniformly to a continuous function y .

$y_{k_n}(x) = \int_0^x f_{k_n}(t, y_{k_n}(t)) dt$, therefore (supposing x is positive since the other way around is the same just with a sign flipped), $\int_0^x f_{k_n}(t, y_{k_n}(t)) dt \rightarrow \int_0^x f(t, y(t)) dt$ by the dominated convergence theorem and continuity of f . The limits on each side must coincide, so $y(x) = \int_0^x f(t, y(t)) dt$. By the fundamental theorem of calculus, y is our solution to the differential equation so we are done at last.

□

Theorem. (Implicit function theorem in 2 dimensions) If f is continuously differentiable in a neighbourhood of a point (x_0, y_0) and $f_y(x, y) \neq 0$ in that neighbourhood, then there exists a unique differentiable function g such that

$$y_0 = g(x_0), f(x, g(x)) = 0$$

in a neighbourhood of x_0 .

Proof. Note that if our curve is $f(x, y) = c$ then we can define a new function equal to $f - c$ so we can assume that 0 is the constant on the right hand side.

On a small rectangle where $f_y \neq 0$, define $\phi(x, y) = \frac{-f_x}{f_y}$. This is continuous, so Peano's theorem guarantees existence of a differentiable solution g of $g' = \phi(x, g)$ with $g(x_0) = y_0$.

Set $h(x) = f(x, g(x))$. Then the chain rule gives $h'(x) = f_x + f_y g'$ which is 0 by the differential equation so h is constant. Since $h(x_0) = f(x_0, y_0) = c$, we have $f(x, g(x)) = c$.

After shrinking to a rectangle, because f_y is continuous and never 0 it must only have one sign there. Therefore, for every fixed x , $y \rightarrow f(x, y)$ is an increasing or decreasing function, and thus $f(x, y)$ determines y if x is determined. Any two local solutions with values in that rectangle must therefore be equal.

□